

Likelihood Construction, Inference for Parametric Survival Distributions

Jih-Chang Yu

September 15, 2025

Introduction

In this section, we obtain the likelihood function for noninformatively right-censored survival data and indicate how to make an inference when a parametric form for the distribution of T is assumed.

- ▶ While the focus of this course is on nonparametric and semiparametric inference, it is useful to consider parametric inference for right-censored survival data for at least two reasons:
 - ▶ First, parametric inference can be very helpful in settings with small sample sizes or when scientific insights can be gleaned from the shapes of survival curves.
 - ▶ Secondly, such inferences provide a useful contrast to inference in more traditional settings, where the observations are direct realizations from the underlying distribution of interest.
- ▶ When observations are subject to censoring, then likelihood methods inform us about the distribution of the observables, which depends on both the underlying survival process and the process that leads to censored observations.

Observables, Likelihood Function

For simplicity, consider the "1-sample" problem in which there are n *i.i.d.* survival times, denoted $T_1, T_2, \dots, T_n$, with a common and unknown c.d.f. $F(\cdot)$, about which we wish to make an inference.

Suppose that we don't observe $T_1, T_2, \dots, T_n$ but instead observe (U_i, δ_i) for $i = 1, 2, \dots, n$, where

$$U_i = \min\{T_i, C_i\}, \quad \delta_i = I(T_i \leq C_i)$$

and C_i is the (fixed or random) "potential censoring time" for item i in the sense that we will have a censored observation at time u if $C_i = u$ and $T_i > u$. Let $\perp$ denote "is independent of". We assume that $T_i \perp C_i$ (noninformative censoring) and that the n pairs (U_i, δ_i) , for $i = 1, \dots, n$, are also *i.i.d.*.

Likelihood Construction

- ▶ Note that the bivariate random variable (U_i, δ_i) consists of a continuous component U_i and a binary component δ_i .
- ▶ (U_i, δ_i) can take two forms:

$$\begin{aligned} (U_i, \delta_i) = (u_i, 1) &: \text{此时 } U_i = T_i \text{ (已经发生事件)} \\ &T_i \text{ is uncensored at time } u_i \\ (U_i, \delta_i) = (u_i, 0) &: \text{此时 } U_i = T_i \text{ (还未发生事件)} \\ &T_i \text{ is censored at time } u_i \end{aligned}$$

- ▶ The likelihood contribution for (U_i, δ_i) , say $L_i(F)$, is proportional to the probability elements corresponding to these two types of outcomes.

Likelihood Construction (Contd.)

$$T_i \sim F \text{ (with pdf } f(\cdot) \text{)}$$

Case 1: C_i known constants

$$P(U_i = u_i, \delta_i)$$

$$L_i(F) = \begin{cases} f(u_i) & \text{if } \delta_i = 1 \\ 1 - F(u_i) & \text{if } \delta_i = 0 \end{cases}$$

for ith individual

$$= (f(u_i))^{\delta_i} (1 - F(u_i))^{1-\delta_i}$$

Therefore,

$$L(F) = \prod_{i=1}^n L_i(F) = \prod_{i=1}^n f(u_i)^{\delta_i} (1 - F(u_i))^{1-\delta_i}$$

Note that (2.1) does not take the form of the usual likelihood function that would result if we had observed $T_1, T_2, \dots, T_n$; however, it functionally depends only on $F(\cdot)$ and thus can be maximized to make inferences about $F(\cdot)$.

Likelihood Construction (Contd.)

Case 2: $C_i \text{ i.i.d. } \sim G$ (G continuous with p.d.f. g) Now suppose that the potential censoring times C_i are independent random variables with distribution function $G(\cdot)$ and density function $g(\cdot)$.

$$P(T_i = u_i, C_i \geq u_i)$$

Event	Also Expressible As	Likelihood Contribution
$U_i = u_i, \delta_i = 1$	$T_i = u_i, C_i \geq u_i$	$f(u_i)[1 - G(u_i)]$
$U_i = u_i, \delta_i = 0$	$T_i \geq u_i, C_i = u_i$	$[1 - F(u_i)] \cdot g(u_i)$

In fact, this is the density of the observables (U_i, δ_i) :

$$\begin{aligned}
 P(U_i \in A, \delta_i = 1) &= P(T_i \in [t, t+\Delta t), C_i \geq T_i) \\
 &= \int_A \int_{c \geq t} f(t) g(c) dc dt = \int_A f(t) [1 - G(t)] dt
 \end{aligned}$$

Likelihood Construction (Contd.)

$$\begin{aligned} P(U_i \in A, \delta_i = 0) &= P(C_i \in A, C_i < T_i) \\ &= \int_{c \in A} \int_{c < t} f(t) g(c) dt dc \\ &= \int_{c \in A} [1 - F(c)] g(c) dc \end{aligned}$$

Note how the independence of T_i and C_i , which ensures noninformative censoring, is instrumental in the determination of the likelihood contributions above. As we see below, the fact that the likelihood contribution factors into one term involving F and another involving G greatly simplifies the resulting inference.

$$\Delta F = \int_t^{t+\Delta t} f(t) dt$$

$$df = P(t < T < t + \Delta t)$$

這是一個很小的區間(概率差)

Likelihood Construction (Contd.)

$$\begin{aligned}
 L(F, G) &= \prod_{i=1}^n \left[f(u_i) (1 - G(u_i))^{\delta_i} \left[(1 - F(u_i)) g(u_i) \right]^{1 - \delta_i} \right] \\
 &= \prod_{i=1}^n \underbrace{\left[f(u_i)^{\delta_i} (1 - F(u_i))^{1 - \delta_i} \right]}_{\substack{\text{if } \delta_i \sim \text{Bernoulli}(\lambda) \\ \text{then } \lambda = \exp(\lambda)}} \left[g(u_i)^{1 - \delta_i} (1 - G(u_i))^{\delta_i} \right]
 \end{aligned}$$

if $\delta_i \sim G(\alpha, 1) \Rightarrow \alpha \neq \lambda(\delta_i)$ function

Suppose that F and G are functionally independent; i.e., $F = F_\theta$, $G = G_\phi$, $\theta \in \Theta$, $\phi \in \Phi$ and par. space is $(\theta, \phi) \in \Theta \times \Phi$. 垂直相交

Then for purposes of inference about F , the terms in L involving G can be regarded as a constant. That is, the maximizing value of $F(\cdot)$ for this likelihood function is the same as from maximizing (2.1). As we illustrate below, this is not to say that the properties of the resulting estimators of $F(\cdot)$ are the same. In this case, the score and information for θ only depend on the first factor, so the usual theory about MLEs can be applied!

Parametric Inference for the Exponential Distribution

Let us examine the use of (1) for the case where we have (noninformatively) right-censored observations from the exponential distribution. We begin with the 1-sample problem and then discuss the comparison of two groups and the analysis of covariates. We assume that the potential censoring times C_i are *i.i.d.* random variables with c.d.f. $G(\cdot)$.

1-Sample Problem

$$f(u_i) = \lambda e^{-\lambda u_i}$$

为T的参数
↑

Suppose $T_1, T_2, \dots, T_n$ are *i.i.d.* $NE(\lambda)$, and subject to noninformative right censoring. Then (2.1) becomes

$$L(F, G) \propto \prod_{i=1}^n f(u_i)^{\delta_i} (1 - F(u_i))^{1 - \delta_i} \Rightarrow \text{定理上}$$

正比关系

$$\begin{aligned} L = L(\lambda) &= \prod_{i=1}^n (\lambda e^{-\lambda u_i})^{\delta_i} (e^{-\lambda u_i})^{1 - \delta_i} \\ &= \lambda^\gamma e^{-\lambda W} \end{aligned}$$

主要推导

where $\gamma = \sum_{i=1}^n \delta_i = \#$ uncensored obs. and $W = \sum_{i=1}^n \underbrace{(u_i)}_{= T_i} =$ total observed time.

1-Sample Problem (Contd.)

$$\begin{aligned}
 \tau = n &\Leftrightarrow \text{all } \delta_i = 1 \\
 &\Leftrightarrow \text{all } u_i = t_i \\
 &\Leftrightarrow \lambda = \frac{1}{\tau} = \frac{1}{\sum t_i / n} \\
 &\quad \downarrow \\
 &\text{MLE of average of } t_i
 \end{aligned}$$

Taking derivatives, we find

$$\begin{aligned}
 \frac{\partial \ln L}{\partial \lambda} &= \frac{\tau}{\lambda} - w \quad ; \quad \frac{\partial^2}{\partial \lambda^2} \ln L = -\frac{\tau}{\lambda^2} \\
 \Rightarrow \hat{\lambda} &= \frac{\tau}{w} \quad ; \quad \hat{I}(\lambda) = \frac{\tau}{\lambda^2}
 \end{aligned}$$

where we use $\hat{I}(\lambda)$ to denote the observed (or sample) information for the sample; e.g., $\hat{I}(\lambda) \stackrel{\text{def}}{=} -\frac{\partial^2 \ln L}{\partial \lambda^2}$.

1-Sample Problem (Contd.)

delta method:

假如 $\hat{\theta} \sim \mathcal{N}(\theta, \sigma^2)$, 存在函数 $g(\theta)$ 连续且可微, 则 $MLE \Rightarrow g(\hat{\theta})$
 $g(\hat{\theta}) \xrightarrow{a.p.} \mathcal{N}(g(\theta), g'(\theta)^2 \sigma^2)$

Note that r , the number of uncensored observations, has the binomial distribution:

$$r \sim \text{Bin}(n, p) \quad , \quad p = P(\delta_i = 1) = \int_0^\infty f(u)(1 - G(u)) du$$

Therefore,

$$I(\lambda) \stackrel{\text{def}}{=} E(I(\lambda)) = \frac{np}{\lambda^2}$$

Notice that to estimate the observed information $\hat{I}(\lambda)$ one does not need to estimate $G(\cdot)$, but $I(\lambda)$ does depend functionally on $G(\cdot)$.

$$g(\hat{\theta}) \cong g(\theta) + (\hat{\theta} - \theta)g'(\theta) + \frac{(\hat{\theta} - \theta)^2}{2}g''(\theta) + \dots$$

$$\Rightarrow \frac{1}{g'(\theta)} [g(\hat{\theta}) - g(\theta)] = \hat{\theta} - \theta \sim \mathcal{N}(0, \sigma^2)$$

$$\Rightarrow g(\hat{\theta}) \stackrel{a.p.}{\sim} \mathcal{N}(g(\theta), g'(\theta)^2 \sigma^2)$$

Asymptotic Results (Contd.)

Assuming the appropriate regularity conditions hold, as $n \rightarrow \infty$,

$$\frac{\hat{\lambda} - \lambda}{\sqrt{I^{-1}(\lambda)}} \xrightarrow{\mathcal{L}} N(0, 1)$$

$\therefore$ or $\hat{\lambda} \sim \mathcal{N}(\lambda, I^{-1}(\lambda))$

(Handwritten notes: A green arrow points from $\hat{\lambda} - \lambda$ to $N(0, 1)$. A red arrow points from $I^{-1}(\lambda)$ to $\frac{\lambda^2}{np}$.)

(see e.g. Andersen et al, Chapter VI, Section 1). Since λ is not known, we can invoke Slutsky's theorem and use the approximation, replacing $\frac{\lambda^2}{np}$ with $\frac{\hat{\lambda}^2}{n\hat{p}}$. That is,

$$\hat{\lambda} \stackrel{\text{apx}}{\sim} N(\lambda, I^{-1}(\lambda)) = N\left(\lambda, \frac{\sigma}{w^2}\right)$$

It turns out that a better approximation is to assume that the log of $\hat{\lambda}$ is normal. Using the *delta method*, this gives:

$$\ln \hat{\lambda} \stackrel{\text{apx}}{\sim} N\left(\ln \lambda, \frac{1}{r}\right)$$

$$\eta(\cdot) = \mathcal{L}(\cdot)$$

Hypothesis Testing and Confidence Intervals

Suppose we wanted to use these results to test the hypothesis $H_0 : \lambda = \lambda_0$ that the true underlying λ equals some specified value, say λ_0 , or to find an approximate 95% confidence interval (CI) for λ . Then under H_0 ,

$$Z_1 \stackrel{\text{def}}{=} \frac{\ell_n(\hat{\lambda}) - \ell_n(\lambda_0)}{\sqrt{1/r}} \approx N(0, 1)$$

→ Use Z_1 to test H_0 .

Hypothesis Testing and Confidence Intervals (Contd.)

Confidence Intervals (CIs):

$$P[\ln \hat{\lambda} - 1.96\sqrt{V_{\hat{\lambda}}} < \ln \lambda < \ln \hat{\lambda} + 1.96\sqrt{V_{\hat{\lambda}}}] \cong 0.95$$

$$\Rightarrow P[\underbrace{\hat{\lambda} \exp(-1.96\sqrt{V_{\hat{\lambda}}})}_{CI_L} < \lambda < \underbrace{\hat{\lambda} \exp(1.96\sqrt{V_{\hat{\lambda}}})}_{CI_U}] \cong 0.95$$

and thus

$$[CI_L, CI_U]$$

is an approximate 95% CI for λ .

AIDS Clinical Trial

Introduction

In this example, we will discuss survival results from an AIDS clinical trial using the STATA data set EX1.dta. Patients were divided into AZT and ddl groups. The data are described in file EX1.doc.

AZT Group Survival Results *Hw*

For the AZT group:

- ▶ Number of patients, $r = 50$
- ▶ Total observation time, $W = 80.8$ years
- ▶ Estimated hazard rate, $\hat{\lambda} = 0.6188$
- ▶ Approximate 95% CI for λ : $(0.4690, 0.8165)$
(based on $\ln \hat{\lambda}$ Normal)

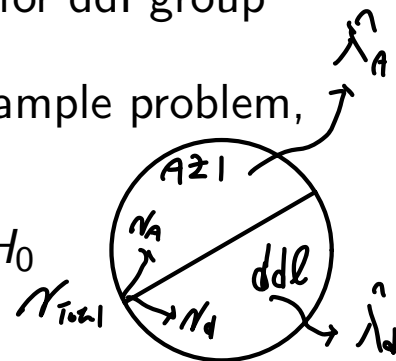
Comparing λ for 2 Groups

Suppose we wanted to compare the λ for 2 groups, say the patients receiving ddl and those receiving AZT.

$$H_0 : \lambda_A = \lambda_d \quad \begin{cases} \lambda_A : & \text{exponential parameter for AZT group} \\ \lambda_d : & \text{exponential parameter for ddl group} \end{cases}$$

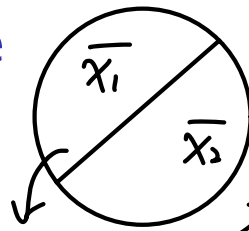
Then using the normal approximations from the 1-sample problem, define

$$Z_2 \stackrel{\text{def}}{=} \frac{\ln \hat{\lambda}_A - \ln \hat{\lambda}_d}{\sqrt{1/r_A + 1/r_d}} \stackrel{\text{apx}}{\sim} N(0, 1) \text{ under } H_0$$



For the data in AZT-ddl.dta, $Z_2 = \dots = 0.218$ ($p=0.83$ (2-sided)) $\rightsquigarrow$ No significant difference between AZT and ddl groups.

Covariates and Inference



$$Y = \alpha + \beta x_i + \varepsilon_i \quad E(\varepsilon_i) = 0 \quad E(Y_i | x_i) = \alpha + \beta x_i$$

Next suppose $\mathbf{z}$ is a $p \times 1$ vector of covariates measured for each subject, and we are interested in assessing whether the components of $\mathbf{z}$ are associated with survival.

For the i^{th} of n independent subjects, suppose that the value of $\mathbf{z}$ is denoted $\mathbf{z}_i$. Thus, the observation for subject i is of the form $(\mathbf{z}_i, U_i, \delta_i)$.

→ How might we analyze such data to make an inference about the association of the components of $\mathbf{z}$ on the underlying survival times?

Approach: Functional Dependence

- **One Approach:**

Continue to assume that each T_i is exponential, but let the parameter λ depend functionally on the corresponding value of $\mathbf{z}$. That is, assume $T_i \sim NE(\lambda_i)$, where $\lambda_i = \text{function}(\mathbf{z}_i)$.

$$E(T_i | \mathbf{z}_i)$$

e.g. $\lambda_i = \lambda_0 \exp(\beta' \mathbf{z}_i)$

$$\Rightarrow \ln \lambda_i = \ln \lambda_0 + \beta' \mathbf{z}_i$$

取ln后存在线性关系

This approach allows us to relate survival times to covariates using a known function. Likelihood methods can be employed to make inferences about the parameters, such as Wald tests.

$$L(F) \propto f(u_i)^{\delta_i} (1 - F(u_i))^{1 - \delta_i}$$

$$= (\lambda_i e^{-\lambda_i u_i})^{\delta_i} (e^{-\lambda_i u_i})^{1 - \delta_i} = \lambda_i^{\delta_i} e^{-\lambda_i u_i}$$

$$= (\lambda_0 e^{\beta' \mathbf{z}_i u_i})^{\delta_i} e^{-\lambda_0 e^{\beta' \mathbf{z}_i u_i} u_i}$$

Feigl & Zelen Approach

$$L(F) = L(\lambda_0, \beta) \quad \lambda_0 \sim L(\lambda_0, \beta) = \ell(\lambda_0, \beta) = \sum_i \ell_i \lambda_0 + \sum_i \beta z_i - \lambda_0 e^{\beta z_i} u_i$$

$$\frac{\partial}{\partial \beta} = \sum_i z_i - \lambda_0 z_i u_i e^{\beta z_i} \stackrel{!}{=} 0 = U_i(\beta)$$

$$\frac{\partial}{\partial \lambda_0} = \frac{\sum_i \ell_i}{\lambda_0} - e^{\beta z_i} u_i = U_i(\lambda_0) \quad L(F) = \prod_{i=1}^n L_i(F)$$

- This approach of functionally relating a parameter of the survival distribution to the covariates was first considered by Feigl & Zelen (1965). $U_i(\lambda_0) = \frac{\sum_{i=1}^n \left[\frac{\ell_i}{\lambda_0} - e^{\beta z_i} u_i \right] \stackrel{!}{=} 0}{\sum_{i=1}^n e^{\beta z_i} u_i}$, $\hat{\lambda}_0 = \frac{\sum_{i=1}^n \ell_i}{\sum_{i=1}^n e^{\beta z_i} u_i}$
- Note: In ordinary regression problems, expressing observations as expected + error is not straightforward in censored data settings.
- In the Feigl & Zelen approach, we express λ_i as a function of the covariate value z_i and consider censoring using the likelihood function.

Select β_* $\Rightarrow \hat{\lambda}(\beta_*)$

Let $U(\beta) = \sum_i z_i - \hat{\lambda}(\beta_*) \sum_i z_i e^{\beta z_i} u_i = 0$

Solve $\hat{\beta}$ such that $U(\hat{\beta}) = 0$

Parametric Inferences for Other Families

More generally, the same approach can be used to make parametric inferences in the presence of (noninformative) right-censored observations for other parametric families (e.g., Weibull, Gamma, ...). One simply uses (2.1) and proceeds in the same way as for the exponential distribution.

One concern with any parametric approach is whether the resulting inference remains valid if the assumed underlying parametric distribution does not fit the data. One way to try to make the resulting inferences robust is to fit a 'weakly structured' parametric model to the data.

Piecewise Exponential Distribution

Consider the 1-sample problem where

$$T_1, T_2, \dots, T_n \text{ i.i.d. } \sim F(\cdot),$$

where $F(\cdot)$ has the "piecewise exponential" distribution. That is, suppose that

$$0 = v_0 < v_1 < \dots < v_k < v_{k+1} \text{ \# } 1$$

is a known partition of $[0, \infty)$, and we assume that the hazard function for T is of the form

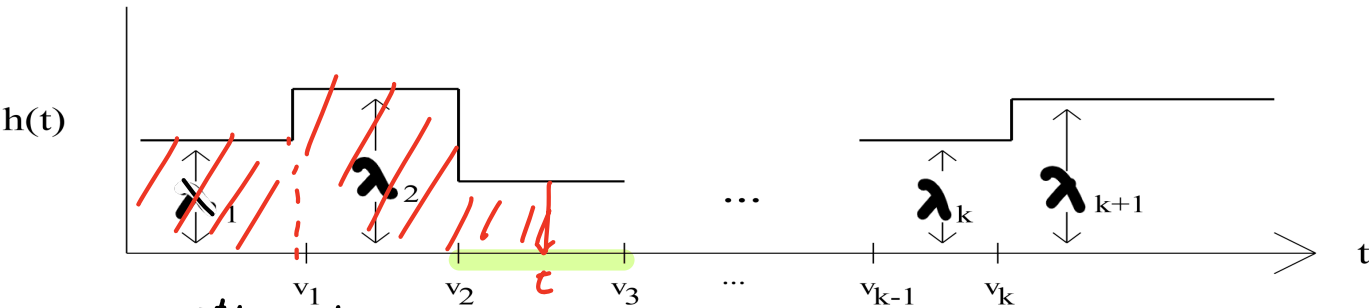
$$h(t) = \lambda_j \text{ for } v_{j-1} < t < v_j$$

Assumed Hazard Function

The unknown parameters of $F(\cdot)$ are $\lambda_1, \dots, \lambda_{k+1}$. The assumed hazard is depicted below:

if $t \in \mathcal{J}$,

$$H(t) = \int_0^t h(\tau) d\tau = (v_1 - v_0) \lambda_1 + (v_2 - v_1) \lambda_2 + (t - v_2) \lambda_3,$$



$$\int_0^t h(\tau) d\tau$$

Let I_j denote the interval $[v_{j-1}, v_j)$, for $j = 1, 2, \dots, k+1$. Then for $t \in I_j$,

$$H(t) = \sum_{\ell=1}^{j-1} \lambda_{\ell} (v_{\ell} - v_{\ell-1}) + \lambda_j (t - v_{j-1})$$

$$F(t) = 1 - e^{-H(t)} \text{ and}$$

$$f(t) = \frac{d}{dt} F(t) = \frac{d}{dt} H(t) e^{-H(t)} = h(t) e^{-H(t)} = \lambda_j e^{-H(t)}$$

Likelihood Function (Part 1)

$$L = \overset{L(F)}{\underset{\downarrow}{L}}(\lambda_1, \lambda_2, \dots, \lambda_{k+1}) = \prod_{i=1}^n f(u_i)^{\delta_i} (1 - F(u_i))^{1-\delta_i}$$

number of parameter

HW

$$= \dots = \prod_{j=1}^{k+1} \lambda_j^{r_j} e^{-\lambda_j w_j}$$

where $r_j = \sum_{i=1}^n \delta_i 1[u_i \in I_j]$ = number of uncensored times in I_j ,
 and $w_j = \sum_{i=1}^n \delta_i \int_{I_j} I(u \leq v) du$ = total time spent in I_j .

$$W_j = \sum_{i=1}^n (u_i - v_{j-1}) I(u_i \in I_j) + (v_j - v_{j-1}) \sum_{i=1}^n I(u_i > v_j)$$

Likelihood Function (Part 2)

Given this likelihood function, we can proceed in the usual way to make inferences. Specifically,

$$\hat{\lambda}_j = \frac{\sigma_j}{w_j}$$

and

$$\hat{I}(\lambda_1, \lambda_2, \dots, \lambda_{k+1}) = \text{diagonal} \left(\frac{r_1}{\lambda_1^2}, \frac{r_2}{\lambda_2^2}, \dots, \frac{r_{k+1}}{\lambda_{k+1}^2} \right)$$

$$\Rightarrow (\hat{\lambda}_1, \hat{\lambda}_2, \dots, \hat{\lambda}_{k+1}) \quad \text{approximately uncorrelated}$$

Approximating $\ln \hat{\lambda}_j$

As before, we can approximate $\ln \hat{\lambda}_j \approx N \left(\ln \lambda_j, \frac{1}{r_j} \right)$.

Applying this to the AZT&ddl.dta (pooling the AZT & ddl groups) gives the following:

$$k = 5, v1 = .25, v2 = .5, v3 = .75, v4 = 1, v5 = 1.5$$

Estimated Parameters

Interval	r_j	W_j	$\hat{\lambda}_j$	$\ln \hat{\lambda}_j$	$\hat{Var}(\ln \hat{\lambda}_j)$
$[0, .25)$	19	42.0	.45	-.79	.05
$[\cdot 25, \cdot 50)$	21	35.8	.59	-.53	.05
$[\cdot 50, \cdot 75)$	28	26.35	1.06	.06	.04
$[\cdot 75, 1.0)$	14	17.86	.78	-.24	.07
$[1.0, 1.5)$	10	21.93	.46	-.79	.10
$[1.5, \infty)$	3	6.90	.43	-.83	.33

Interpretation of Results (Part 1)

Taken at face value, the plot suggests that perhaps the hazard function is unimodal. However, the variances of the $\hat{\lambda}_j$ are considerable and not equivalent.

One could use a weakly structured model such as this to assess whether a specific parametric assumption (say, the Weibull distribution) is reasonable, or as an estimator in its own right.

Interpretation of Results (Part 2)

NOTE 1: In the preceding, it may be preferable to plot $\ln \hat{\lambda}_j$ (or equivalently, $\hat{\lambda}_j$ on a log scale) instead of $\hat{\lambda}_j$ because the precision will vary less with variations in the λ_j .

NOTE 2: The preceding assumes the intervals were pre-specified. An alternative is to let the data determine the intervals so that there are equal numbers of uncensored events (e.g., 5) per interval. The resulting estimates of $\ln \hat{\lambda}_j$ will then have approximately equal precision. See Cox (1979) for details.

Semiparametric Approaches (Part 1)

- ▶ Another approach: Allow both λ_j and v_j to be random.
- ▶ Special case: $k = 1$, partition time axis into 2 intervals.
- ▶ Referred to as "changepoint" problem.
- ▶ Postulate a distribution shift at unknown time v_1 .

Semiparametric Approaches (Part 2)

- ▶ Unknowns: Hazards λ_1 and λ_2 , time v_1 of shift.
- ▶ Traditional estimators may give poor results.
- ▶ Example: Hinkley (1970) discusses such problems.

Unrestricted Likelihood Function (Part 1)

- ▶ Another approach would be to allow both the λ_j and the v_j to be random.
- ▶ The special case where $k = 1$ and we partition the time axis into 2 intervals is sometimes referred to as a "changepoint" problem: we have a process whose distribution we postulate to "shift" at some unknown time point v_1 .
- ▶ The unknowns in our setting are the hazards λ_1 and λ_2 and the time, v_1 at which the shift occurs.
- ▶ See, for example, Hinkley (1970) for an example of how such problems arise and how traditional estimators can sometimes give poor results.

Unrestricted Likelihood Function (Part 2)

- ▶ Recall equation (2.2), the likelihood function $L(F, G)$ when the potential censoring times are assumed to be i.i.d. with c.d.f. $G(\cdot)$.
- ▶ In the development following (2.2) we assumed that $G(\cdot)$ was from some parametric family, and thus $L(F, G)$ was actually a function over some finite-dimensional space.
- ▶ Suppose instead we wanted to leave $G(\cdot)$ unrestricted; conceptually, we still can think about the maximizing value, say $(\hat{F}, \hat{G})$, of (2.2), where the space over which the maximum is taken involves both the parametric space for $F(\cdot)$ and the unrestricted space for $G(\cdot)$.

Unrestricted Likelihood Function (Part 3)

- ▶ What can we say about the solution $(\hat{F}, \hat{G})$? Given the factorization of (2.2) as a product of two terms, one involving F and one involving G , it follows that $\hat{F}$ can be obtained by simply maximizing (2.1).
- ▶ Another way to view this is to consider the "profile likelihood function" for F , defined as $L_p(F) = \sup_G L(F, G)$. This can be shown to be proportional to (2.1).

Unrestricted Likelihood Function (Part 4)

- ▶ A formal theory for inference in such a "semiparametric" setting exists.